

Izveštaj o Testiranju Opterećenja Sistema

Factory Management Module

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Predmet: Napredne veb tehnologije

1. Specifikacija Test Okruženja

1.1 Hardverska Konfiguracija

Komponenta	Specifikacija
Procesor	[AMD Ryzen 5 5600 @ 3.5GHz]
RAM Memorija	[16 GB DDR4 3200MHz]
Disk	[SSD 1TB NVMe]
Operativni Sistem	Windows 11

1.2 Softverska Konfiguracija

Komponenta	Verzija	Port
Spring Boot Backend	3.x	8080
PostgreSQL	15.x	5432
Redis (Docker)	7.x	6379
InfluxDB (Docker)	2.7	8086
RabbitMQ (Docker)	3.x	5672, 15672

1.3 Količina Test Podataka

Entitet	Broj Zapisa
Factories (Fabrike)	[20,000]
Managers (Korisnici)	[1,000,000]
InfluxDB telemetrija	[26,298,000]

1.4 Alat za Testiranje

- Locust v2.x - Python-based load testing framework
- Web UI na <http://localhost:8089>
 - Konfiguracija: wait_time between(3, 8) sekundi

2. Testirani Endpoint-ovi

Pregled 10 Testiranih Endpoint-ova

#	Endpoint	HTTP	Opis
1	/api/v1/factories/paged	GET	Paginirana lista fabrika
2	/api/v1/factories/search/paged	GET	Pretraga fabrika
3	/api/v1/factories/{id}	GET	Detalji fabrike
4	/api/v1/factories/filter	GET	Filtriranje fabrika
5	/api/v1/products/paged	GET	Paginirana lista proizvoda
6	/api/v1/products/search	GET	Pretraga proizvoda
7	/api/v1/products/{id}	GET	Detalji proizvoda
8	/api/v1/managers/paged	GET	Paginirana lista menadžera
9	/api/v1/managers/search	GET	Pretraga menadžera
10	/api/v1/managers	POST	Kreiranje menadžera

3. Rezultati Testiranja

3.1 Test sa 10 Korisnika

Spawn rate: 10 user/sec

Endpoint	# Requests	# Fails	Median (ms)	95%ile (ms)	Avg (ms)
GET /factories/paged	15	0	22	42	23.83
GET /factories/search/paged	13	0	41	66	41.19
GET /factories/{id}	17	0	6	20	7
GET /factories/filter	4	0	10	58	28.37
GET /products/paged	19	0	9	3100	176.17
GET /products/search	15	0	9	160	22.49
GET /products/{id}	18	0	180	340	195.4
GET /managers/paged	8	0	120	160	126.49
GET /managers/search	4	0	500	1400	717.18
POST /managers	7	0	83	180	97.7
UKUPNO	206	0	14	390	104.89

3.2 Test sa 50 Korisnika

Spawn rate: 10 users/sec

Endpoint	# Requests	# Fails	Median (ms)	95%ile (ms)	Avg (ms)
GET /factories/paged	85	0	18	38	20.69

GET /factories/search/paged	64	0	44	73	47.69
Endpoint	#	#	Median	95%ile	Avg
GET /factories/{id}	Requests	Fails	(ms)	(ms)	(ms)
	113	0	4	9	4.69
GET /factories/filter	52	0	9	47	13.29
GET /products/paged	87	0	6	32	51.96
GET /products/search	42	0	12	25	7.38
GET /products/{id}	122	0	43	250	70.76
GET /managers/paged	61	0	120	730	211.73
GET /managers/search	30	0	540	2100	869.45
POST /managers	26	0	79	110	82.99
UKUPNO	1025	0	13	470	77.15

3.3 Test sa 100 Korisnika

Spawn rate: 10 users/sec

Endpoint	# Requests	# Fails	Median (ms)	95%ile (ms)	Avg (ms)
GET /factories/paged	117	0	56	5200	1289.15
GET /factories/search/paged	90	0	98	5400	1355.14
GET /factories/{id}	195	0	8	2600	320.59
GET /factories/filter	95	0	20	3400	475.11
GET /products/paged	125	0	44	5100	1243.21
GET /products/search	81	0	75	5200	1434.26
GET /products/{id}	165	0	190	3500	970.99
GET /managers/paged	75	0	590	3500	1007.52
GET /managers/search	45	0	1500	6200	2439.97
POST /managers	44	0	120	3400	592.22
UKUPNO	1595	0	65	4300	944.1

3.5 Test sa 2000 Korisnika

Spawn rate: 50 users/sec

Endpoint	# Requests	# Fails	Median (ms)	95%ile (ms)	Avg (ms)
GET /factories/paged	301	0	990	12000	3925.17
GET /factories/search/paged	206	0	520	12000	3620.82
GET /factories/{id}	588	0	6	3300	268.35
GET /factories/filter	238	0	23	2600	257.84
GET /products/paged	296	0	2000	12000	4391.95

Endpoint	# Requests	# Fails	Median (ms)	95%ile (ms)	Avg (ms)
GET /products/search	421	0	2300	10000	3602.3
GET /managers/paged	244	0	770	5000	1381.75
GET /managers/search	101	0	3300	8900	3560.6
POST /managers	146	0	110	3400	368.72
UKUPNO	4336	0	100	10000	2216.85

3.5 Test sa 10000 Korisnika

Spawn rate: 50 users/sec

Endpoint	# Requests	# Fails	Median (ms)	95%ile (ms)	Avg (ms)
GET /factories/paged	565	0	26000	55000	29330.95
GET /factories/search/paged	347	0	24000	53000	26890.97
GET /factories/{id}	762	0	18000	47000	20494.52
GET /factories/filter	386	0	20000	47000	20871.34
GET /products/paged	519	0	30000	56000	30777.85
GET /products/search	363	0	31000	56000	31497.47
GET /products/{id}	771	0	26000	52000	26655.59
GET /managers/paged	385	0	22000	48000	21868.44
GET /managers/search	219	0	22000	47000	22469.93
POST /managers	182	0	16000	46000	17101.36.72
UKUPNO	7493	0	2400	52000	24792.29

4. Predlozi za Dalje Unapređenje

4.1 Kratkoročni (Moguće Implementirati)

#	Predlog	Očekivano Poboljšanje
1	Dodavanje složenih indeksa	30-50% brži upiti
2	Connection pooling optimizacija	Bolja propusnost

4.2 Dugoročni (Zahtevaju Više Resursa)

#	Predlog	Opis
1	Horizontalno skaliranje	Load balancer + više backend instanci
2	Database replikacija	Read replicas za SELECT upite

8. Zaključak

Testiranje opterećenja sistema pokazalo je da aplikacija stabilno radi sa do 10000 istovremenih korisnika. Sa povećanjem broja korisnika, performanse se degradiraju, što je očekivano ponašanje za sistem pokrenut na jednoj mašini sa ograničenim resursima.

Glavni bottleneck identifikovan je u upitima nad tabelom `users` koja sadrži 3 miliona zapisa. Dodavanjem database indeksa i Redis keširanja, performanse su poboljšane.

Za produkciono okruženje preporučuje se:

- Distribuirana arhitektura sa više servera
- Database replikacija
- Povećani hardverski resursi